

Ana Lorena Ortiz Loyola

Frankfurt am Main, Germany

☎ +49 176 62145797 • ✉ a01750283@tec.mx • LinkedIn

Data science student (Tecnológico de Monterrey, expected 2027) with hands-on experience turning raw data into reliable, decision-ready reporting at Deutsche Bundesbank (AnaCredit): data cleaning, structured analysis, and dashboard-based communication for both technical and business stakeholders. Comfortable applying AI tools to accelerate analysis and research tasks, and motivated to support data-driven decision-making for marketing and sales teams.

Core Competencies

- **Data Analysis & Market Insights:** Structured analysis of large, real-world datasets to identify trends and inform decisions
- **BI & Visualization:** Building dashboards and structured reporting (Tableau, Matplotlib, Seaborn); comfortable ramping up quickly on new BI tools such as Power BI
- **Programming:** Python (Pandas, NumPy, Scikit-Learn), SQL; data cleaning and transformation
- **AI Tools for Efficiency:** Regular use of Claude Code as an AI coding/research assistant to accelerate analysis and documentation tasks
- **Machine Learning:** K-Means clustering, Random Forest, SVM; predictive modeling on large real-world datasets
- **Collaboration & Communication:** Stakeholder-facing requirement translation, clear communication in German and English, Jira, Git

Professional Experience

- **Deutsche Bundesbank (AnaCredit)** **Frankfurt am Main, Germany**
Data Engineer / Intern 04/2026–07/2026
 - Developed automated migration tests (row counts, referential integrity, aggregate consistency) using Pandas DataFrames, verifying that new tooling reproduced identical business logic before roll-out - an ETL testing and quality-assurance workflow
 - Implemented automated error detection identifying deviations (duplicate IDs, value-range violations) in reporting data, then structured and documented findings for review
 - Automated Jira ticket creation and emails to reporting banks, translating technical findings into clear, actionable communication for non-technical stakeholders
 - Significantly reduced response time to data quality issues through end-to-end ownership, from detection to resolution

Projects

- **Tecnológico de Monterrey**
Predictive Modeling – Healthcare Optimization 10/2024–12/2024
 - Analyzed 25+ years of Mexican national health data to identify patient risk profiles and resource bottlenecks
 - Built predictive models (Random Forest, Neural Networks) and K-Means clustering to segment high-risk groups, directly analogous to clustering structured records into meaningful categories

- **Personal Project (GitHub)**
Banking Data Pipeline – Data Cleaning & Audit Log 05/2026
 - Built an end-to-end pipeline for simulated bank transactions (150 accounts) with a normalized SQLite database and foreign-key integrity checks
 - Staged data cleaning with a fully auditable log; automated balance maintenance via SQL triggers; built using **Claude Code** as an AI coding assistant to accelerate implementation
- **Tecnológico de Monterrey**
Lattice-Based Cryptography Research 10/2024–02/2025
 - Implemented and analyzed the complexity of lattice-based cryptographic algorithms (LWE, NTRU) in Python
 - Contributed to a scientific publication analyzing security parameters and runtime behavior across lattice dimensions

Education

- **Tecnológico de Monterrey** **Mexico**
BSc in Data Science 2022–2027
Grade: 1.3 (Academic Excellence Scholarship; DAAD KOSPIE Scholarship). Topics: data analysis & statistics, machine learning & AI, cryptography & security, NLP, numerical optimization.
- **Universität Göttingen** **Germany**
Exchange Semester 2025–2026
- **Tecnológico de Monterrey** **Mexico**
Abitur 2019–2022
Grade: 1.0. National Physics Olympiad 2020; Honorable Mention, Physics Olympiad 2021.

Languages

- Spanish (native), English (C1, TOEFL iBT 105/120), German (C1).

Publications

- Contributed to a scientific publication on lattice-based cryptography (LWE, NTRU security parameter and runtime analysis), with Prof. A. F. De Abiega L'Eglise, Tecnológico de Monterrey (citation pending publication).

Honors and Awards

- **Academic Excellence Scholarship** – Tecnológico de Monterrey.
- **DAAD KOSPIE Scholarship**.
- **National Physics Olympiad** – Mexico (2020).
- **Honorable Mention, Physics Olympiad** (2021).

References

- Available upon request.